

- 2 (a) Homeostasis allows mammals to maintain relatively constant internal conditions despite changes in the external environment. One condition that is controlled is the water potential of the blood.

- (i) Complete the list to show the receptors, coordination systems, effectors and response that achieve the homeostatic control of the water potential of the blood. Two of the answers have been started for you.

stimulus: decrease in blood water potential

initial receptors:

coordination systems: nervous system and

final effectors: distal convoluted tubule and

response: [4]

- (ii) Explain why homeostasis of the water potential of the blood is necessary.

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..... [2]

(b) The glomerular filtration rate of the kidneys is $200 \text{ dm}^3 \text{ day}^{-1}$.

The mean concentration of glucose in the blood is 5 mmol dm^{-3} .

1 mmol of glucose weighs 0.18 g.

(i) Calculate the mass of glucose that is filtered from the blood by the kidneys in one day.

.....g [1]

(ii) A healthy person does **not** lose the mass of glucose calculated in **2(b)(i)** in urine each day due to the processes occurring in the kidneys.

Describe the processes that occur in the kidneys to prevent the loss of this glucose in urine.

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..... [4]

[Total: 11]